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FACILITATING COMMUNICATION BETWEEN COMPUTING PLATFORMS

Abstract

Facilitating communication between computing platforms, including establishing, via a game agnostic communication service (GACS), communication with a first and a second computing platform; receiving i) first game state data associated with a first video game application executing at the first computing platform and ii) second game state data associated with a second video game application executing at the second computing platform; receiving a request to display at the first computing platform a first communication interface of the GACS; generating content for the first communication interface based at least in part on the first or the second game state data; causing to display the generated content via the first communication interface; receiving selection of one of the communication messages to transmit to the second computing platform; transmitting the selected communication message to the second computing platform for display via a second communication interface of the GACS at the second computing platform.

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Background/Summary

BACKGROUND
Field of the Disclosure
[0001] The disclosure relates generally to computing platforms, and in particular, facilitating communication between computing platforms.
Description of the Related Art
[0002] Some video games are designed to enable multiple players to play an instance of the video game together. Many of these multiplayer video games include functionality that enables the players to communicate with each other while they play the video game. Often, this communication takes the form of textual communication, which may be referred to as chat. For instance, one player may type a message using a keyboard and send this message to a particular player or to all other players who are playing the video game with the player that wrote the message.
SUMMARY
[0003] Innovative aspects of the subject matter described in this specification may be embodied in a method of facilitating communication between computing platforms, including establishing, over a network and via a game agnostic communication service, communication with a first computing platform and a second computing platform; receiving first game state data associated with a first video game application executing at the first computing platform; receiving second game state data associated with a second video game application executing at the second computing platform; receiving a request to display at the first computing platform a first communication interface of the game agnostic communication service, the first communication interface including one or more interactable graphical elements; generating content for the first communication interface to be displayed at the first computing platform, the content being generated based at least in part on the first game state data or the second game state data, wherein the content to be displayed includes communication messages corresponding to each of the one or more interactable graphical elements, wherein each of the communication messages comprise a function associated with the first video game application or the second video game application; causing to display the generated content via the first communication interface of the first computing platform; receiving a selection of one of the communication messages displayed via the first communication interface of the first computing platform to transmit to the second computing platform; and transmitting the selected communication message to the second computing platform for display via a second communication interface of the game agnostic communication service at the second computing platform.
[0004] Other embodiments of these aspects include corresponding systems, apparatus, and computer programs, configured to perform the actions of the methods, encoded on computer storage devices.
[0005] These and other embodiments may each optionally include one or more of the following features. For instance, generating the content to be displayed is based at least in part on the type of platforms of the first computing platform and second computing platform. Both the first video game application and the second video game application are of the same video game application. The first video game application is of a first type of video game application and the second video game application is of a second type of video game application differing from the first type of video game application. Translating the selected communication message from a first format associated with the first computing platform to a second format associated with the second computing platform; and displaying the translated selected communication message via the second communication interface of the second computing platform. Translating the selected

communication message includes translating the selected communication message based on one or more user settings of the second computing platform. The user settings include accessibility settings, language settings, or both. Generating the content to be displayed is also based on one or more settings from at least the first video game application or the second video game application. The first computing platform is of a first type of gaming platform and the second computing platform is of a second type of gaming platform, the first type of gaming platform differing from the second type of gaming platform. Instantiating the first communication interface at the first computing platform by the first video game application.
[0006] The details of one or more embodiments of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other potential features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0007] FIG. **1** is a block diagram of selected elements of an embodiment of a computing platform.
[0008] FIGS. **2** and **3** illustrate respective block diagrams of a computing environment including multiple computing platforms, game platform servers, and a game services server.
[0009] FIGS. **4**A-**4**C illustrate respective graphical user interfaces at a computing platform.
[0010] FIGS. **5**A-**5**C illustrate respective graphical user interfaces at a computing platform.
[0011] FIG. **6** illustrates a graphical user interface at a computing platform.
[0012] FIG. **7** illustrates a method for facilitating communication between computing platforms.
DESCRIPTION OF PARTICULAR EMBODIMENT(S)
[0013] This disclosure discusses methods and systems for facilitating communication between computing platforms. In short, messaging between computing platforms can include “quick messaging systems” that can provide efficient communication. The “quick messaging system” can be i) agnostic to any application (e.g., game) executing at the computing platforms and ii) platform agnostic. Additionally, the “quick messaging system” can leverage data and functions (state and telemetry data) from the applications and platforms to streamline communication functions, described further herein.
[0014] Specifically, this disclosure discusses a system and a method for establishing, over a network and via a game agnostic communication service, communication with a first computing platform and a second computing platform; receiving first game state data associated with a first video game application executing at the first computing platform; receiving second game state data associated with a second video game application executing at the second computing platform; receiving a request to display at the first computing platform a first communication interface of the game agnostic communication service, the first communication interface including one or more interactable graphical elements; generating content for the first communication interface to be displayed at the first computing platform, the content being generated based at least in part on the first game state data or the second game state data, wherein the content to be displayed includes communication messages corresponding to each of the one or more interactable graphical elements, wherein each of the communication messages comprise a function associated with the first video game application or the second video game application; causing to display the generated content via the first communication interface of the first computing platform; receiving a selection of one of the communication messages displayed via the first communication interface of the first computing platform to transmit to the second computing platform; and transmitting the selected communication message to the second computing platform for display via a second communication interface of the game agnostic communication service at the second computing platform.

[0015] In the following description, details are set forth by way of example to facilitate discussion of the disclosed subject matter. It should be apparent to a person of ordinary skill in the field, however, that the disclosed embodiments are exemplary and not exhaustive of all possible embodiments.

[0016] For the purposes of this disclosure, a computing platform may include an instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize various forms of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, a computing platform may be a personal computer, a PDA, a consumer electronic device, a network storage device, or another suitable device and may vary in size, shape, performance, functionality, and price. The computing platform may include memory, one or more processing resources such as a central processing unit (CPU) or hardware or software control logic. Additional components of the computing platform may include one or more storage devices, one or more communications ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, and a video display. The computing platform may also include one or more buses operable to transmit communication between the various hardware components.

[0017] For the purposes of this disclosure, computer-readable media may include an instrumentality or aggregation of instrumentalities that may retain data and/or instructions for a period of time. Computer-readable media may include, without limitation, storage media such as a direct access storage device (e.g., a hard disk drive or floppy disk), a sequential access storage device (e.g., a tape disk drive), compact disk, CD-ROM, DVD, random access memory (RAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), and/or flash memory (SSD); as well as communications media such as wires, optical fibers, microwaves, radio waves, and other electromagnetic and/or optical carriers; and/or any combination of the foregoing.

[0018] Particular embodiments are best understood by reference to FIGS. **1-7** wherein like numbers are used to indicate like and corresponding parts.

[0019] Turning now to the drawings, FIG. **1** illustrates a block diagram depicting selected elements of a computing platform **100** in accordance with some embodiments of the present disclosure. In various embodiments, computing platform **100** may represent different types of portable computing platforms, such as, display devices, head mounted displays, head mount display systems, smart phones, tablet computers, notebook computers, media players, digital cameras, 2-in-1 tablet-laptop combination computers, and wireless organizers, or other types of portable computing platforms. In one or more embodiments, computing platform **100** may also represent other types of computing platforms, including desktop computers, server systems, controllers, and microcontroller units, among other types of computing platforms. Components of computing platform **100** may include, but are not limited to, a processor subsystem **120**, which may comprise one or more processors, and system bus **121** that communicatively couples various system components to processor subsystem **120** including, for example, a memory subsystem **130**, an I/O subsystem **140**, a local storage resource **150**, and a network interface **160**. System bus **121** may represent a variety of suitable types of bus structures, e.g., a memory bus, a peripheral bus, or a local bus using various bus architectures in selected embodiments. For example, such architectures may include, but are not limited to, Micro Channel Architecture (MCA) bus, Industry Standard Architecture (ISA) bus, Enhanced ISA (EISA) bus, Peripheral Component Interconnect (PCI) bus, PCI-Express bus, HyperTransport (HT) bus, and Video Electronics Standards Association (VESA) local bus.

[0020] As depicted in FIG. **1**, processor subsystem **120** may comprise a system, device, or apparatus operable to interpret and/or execute program instructions and/or process data, and may include a microprocessor, microcontroller, digital signal processor (DSP), application specific integrated circuit (ASIC), or another digital or analog circuitry configured to interpret and/or execute program instructions and/or process data. In some embodiments, processor subsystem **120** may interpret and/or execute program instructions and/or process data stored locally (e.g., in memory subsystem **130** and/or another component of the computing platform). In the same or alternative embodiments, processor subsystem **120** may interpret and/or execute program instructions and/or process data stored remotely (e.g., in network storage resource **170**).

[0021] Also in FIG. **1**, memory subsystem **130** may comprise a system, device, or apparatus operable to retain and/or retrieve program instructions and/or data for a period of time (e.g., computer-readable media). Memory subsystem **130** may comprise random access memory (RAM), electrically erasable programmable read-only memory (EEPROM), a PCMCIA card, flash memory, magnetic storage, opto-magnetic storage, and/or a suitable selection and/or array of volatile or non-volatile memory that retains data after power to its associated computing platform, such as system **100**, is powered down.

[0022] In computing platform **100**, I/O subsystem **140** may comprise a system, device, or apparatus generally operable to receive and/or transmit data to/from/within computing platform **100**. I/O subsystem **140** may represent, for example, a variety of communication interfaces, graphics interfaces, video interfaces, user input interfaces, and/or peripheral interfaces. In various embodiments, I/O subsystem **140** may be used to support various peripheral devices, such as a touch panel, a display adapter, a keyboard, an accelerometer, a touch pad, a gyroscope, an IR sensor, a microphone, a sensor, or a camera, or another type of peripheral device.

[0023] Local storage resource **150** may comprise computer-readable media (e.g., hard disk drive, floppy disk drive, CD-ROM, and/or other type of rotating storage media, flash memory, EEPROM, and/or another type of solid state storage media) and may be generally operable to store instructions and/or data. Likewise, the network storage resource may comprise computer-readable media (e.g., hard disk drive, floppy disk drive, CD-ROM, and/or other type of rotating storage media, flash memory, EEPROM, and/or other type of solid state storage media) and may be generally operable to store instructions and/or data.

[0024] In FIG. **1**, network interface **160** may be a suitable system, apparatus, or device operable to serve as an interface between computing platform **100** and a network **110**. Network interface **160** may enable computing platform **100** to communicate over network **110** using a suitable transmission protocol and/or standard, including, but not limited to, transmission protocols and/or standards enumerated below with respect to the discussion of network **110**. In some embodiments, network interface **160** may be communicatively coupled via network **110** to a network storage resource **170**. Network **110** may be a public network or a private (e.g., corporate) network. The network may be implemented as, or may be a part of, a storage area network (SAN), personal area network (PAN), local area network (LAN), a metropolitan area network (MAN), a wide area network (WAN), a wireless local area network (WLAN), a virtual private network (VPN), an intranet, the Internet or another appropriate architecture or system that facilitates the communication of signals, data and/or messages (generally referred to as data). Network interface **160** may enable wired and/or wireless communications (e.g., NFC or Bluetooth) to and/or from computing platform **100**.

[0025] In particular embodiments, network **110** may include one or more routers for routing data between client computing platforms **100** and server computing platforms **100**. A device (e.g., a client computing platform **100** or a server computing platform **100**) on network **110** may be addressed by a corresponding network address including, for example, an Internet protocol (IP) address, an Internet name, a Windows Internet name service (WINS) name, a domain name or other system name. In particular embodiments, network **110** may include one or more logical groupings of network devices such as, for example, one or more sites (e.g., customer sites) or subnets. As an example, a corporate network may include potentially thousands of offices or branches, each with its own subnet (or multiple subnets) having many devices. One or more client computing platforms **100** may communicate with one or more server computing platforms **100** via any suitable connection including, for example, a modem connection, a LAN connection including the Ethernet or a broadband WAN connection including DSL, Cable, Ti, T3, Fiber Optics, Wi-Fi, or a mobile network connection including GSM, GPRS, 3G, or WiMax.

[0026] Network **110** may transmit data using a desired storage and/or communication protocol, including, but not limited to, Fibre Channel, Frame Relay, Asynchronous Transfer Mode (ATM), Internet protocol (IP), other packet-based protocol, small computer system interface (SCSI), Internet SCSI (iSCSI), Serial Attached SCSI (SAS) or another transport that operates with the SCSI protocol, advanced technology attachment (ATA), serial ATA (SATA), advanced technology attachment packet interface (ATAPI), serial storage architecture (SSA), integrated drive electronics (IDE), and/or any combination thereof. Network **110** and its various components may be implemented using hardware, software, or any combination thereof.

[0027] Turning to FIG. **2**, FIG. **2** illustrates an environment **200** including a first computing platform **202**, a second computing platform **204**, a first game platform server **206**, a second game platform server **208**, and a game services server **210**. In some examples, the computing platforms **202**, **204** are each similar to, or include, the computing platform **100** of FIG. **1**. In some examples, the game platform servers **206**, **208** and the game services server **210** are similar to, or include, the computing platform **100** of FIG. **1**. The environment **200** can include any number of computing platforms, e.g., two or more computing platforms. The environment **200** can include any number of game services servers **210**.

[0028] The computing platforms **202**, **204** can be in communication with the game platform servers **206**, **208**, respectively, and the game services server **210** over a network **212** (e.g., the "cloud" or the Internet). In some examples, the game platform servers **206**, **208** are in communication with the game services server **210**.

[0029] In some examples, the first computing platform **202** is the same as the second computing platform **204**. That is, a type (model, or manufacturer) of platform of the first computing platform **202** is the same as the type (model, or manufacturer) of platform of the second computing platform **204**. In some examples, the first computing platform **202** differs from the second computing platform **204**. That is, a type (model, or manufacturer) of platform of the first computing platform **202** differs from the type (model, or manufacturer) of platform of the second computing platform **204**.

[0030] In short, messaging between the computing platforms **202**, **204** via the game services server **210** can include "quick messaging systems" that can provide efficient communication. The "quick messaging system" can be i) agnostic to any application (e.g., game) executing at the computing platforms and ii) platform agnostic. Additionally, the "quick messaging system" can leverage data and functions (state and telemetry data) from the applications and platforms to streamline communication functions, described further herein.

[0031] FIG. **3** illustrates a detailed view of the environment **200**. Specifically, the first computing platform **202** can include a first video game application **302** and a graphical user interface (GUI) **308**. In some examples, the first computing platform **202** is a gaming platform (a first type of a gaming platform—home video console, portable video device, handheld video device, smartphone, or a personal computing device).

[0032] A user **380** can interact with the first computing platform **202**. For example, the user **380** can interact with the first computing platform **202** via a user device **382**. The user device **382** can include any user interactable device that can provide input to the first computing platform **202** (e.g., wired or wireless). For example, the user device **382** is any device that provides tactile input to the first computing platform **202**. For example, the user device **382** can include a controller (video game controller).

[0033] The second computing platform **204** can include a second video game application **352** and a graphical user interface (GUI) **358**. In some examples, the second computing platform **204** is a gaming platform (a second type of a gaming platform—home video console, portable video device, handheld video device, smartphone, or a personal computing device). In some examples, the second type of the gaming platform of the second computing platform **204** can be different than the first type of the gaming platform of the first computing platform **202**. In some examples, the second type of the gaming platform of the second computing platform **204** can be the same, or similar, to the first type of the gaming platform of the first computing platform **202**.

[0034] A user **390** can interact with the second computing platform **204**. For example, the user **390** can interact with the second computing platform **204** via a user device **392**. The user device **392** can include any user interactable device that can provide input to the second computing platform **204** (e.g., wired or wireless). For example, the user device **392** is any device that provides tactile input to the second computing platform **204**. For example, the user device **392** can include a controller (video game controller). In some examples, the user device **392** is the same as the user device **382** (a type of the user device **392** is the same as a type of the user device **382**). In some examples, the user device **392** differs from the user device **382** (a type of the user device **392** differs from a type of the user device **382**).

[0035] In some examples, the first video game application **302** and the second video game application **352** are the same, or substantially the same, video game applications (or online video game applications). That is, the first video game application **302** and the second video game application **352** are the same with the first video game application **302** being deployed at the first computing platform **202** and the second video game application **352** being deployed at the second computing platform **204**.

[0036] In some examples, the first video game application **302** and the second video game application **352** are different video game applications (or online video game applications), or differing types of video game applications (or online video game applications). That is, the first video game application **302** and the second video game application **352** are different types with the first video game application **302** being deployed at the first computing platform **202** and the second video game application **352** being deployed at the second computing platform **204**.

[0037] The first computing platform **202** can be in communication with the first game platform server **206**. The first game platform server **206** can facilitate execution of the first video game application **302** at the first computing platform **202**. For example, the first game platform server **206** can perform such functions as authentication of the user **380**, including determining whether the user **380** is associated with a valid user account, a status of such user account (e.g., active, or banned), permissions associated with the user account, and the like. For example, the first game platform server **206** can perform such functions as player networking to identify social online relationships with other users. For example, the first game server platform **206** can perform such functions as matchmaking to group together users during group play of the first video game application **302**. For example, the first game platform server **206** can perform such functions as game management to host the online execution of the first video game application **302**.

[0038] The second computing platform **204** can be in communication with the second game platform server **208**. The second game platform server **208** can facilitate execution of the second video game application **352** at the second computing platform **304**. For example, the second game platform server **208** can perform such functions as authentication of the user **390**, including determining whether the user **390** is associated with a valid user account, a status of such user account (e.g., active, or banned), permissions associated with the user account, and the like. For example, the second game platform server **208** can perform such functions as player networking to identify social online relationships with other users. For example, the second game platform server **208** can perform such functions as matchmaking to group together users during play of the second video game application **352**. For example, the second game platform server **208** can perform such functions as game management to host the online execution of the second video game application **352**.

[0039] The game services server **210** can be in communication with the first computing platform

202 and the second computing platform **204**. The game services server **210** can include a game agnostic communication service (GACS) **304**. The GACS **304** can facilitate communication between the first computing platform **202** and the second computing platform **204**, described further herein. In some examples, the GACS **304** is agnostic to the first video game application **302** and the second video game application **352**. That is, the GACS **304** is not specific or correlated to the first video game application **302** or the second video game application **352**. The GACS **304** is independent of the first video game application **302** and the second video game application **352**, and independent of any particular parameters of the first video game application **302** and the second video game application **352**. Furthermore, the GACS **304** is independent of the first computing platform **202** and the second computing platform **204**; for example, the GACS **304** is independent of the type of the first computing platform **202** and the second computing platform **204**, and/or the technology implemented at the first computing platform **202** and the second computing platform **204**.

[0040] In some examples, the game services server **210** can represent multiple servers—a distributed computing platform. The GACS **304** can be distributed across any number of game services servers **210**.

[0041] To that end, the GACS **304** can facilitate communication between the first computing platform **202** and the second computing platform **204**, described further herein. Specifically, communication is established (e.g., over the network **212** shown in FIG. **1**) between i) the game services server **210** and the first computing platform **202** and between ii) the game services server **210** and the second computing platform **204**. Specifically, communication is established between the game services server **210** and the computing platforms **202**, **204** via the GACS **204**.

[0042] The game services server **210**, and specifically, the GACS **204**, can receive first game state data **306** from the first computing platform **202**. In particular, the first video game application **302** can be executing at the first computing platform **202** (e.g., a user **380** is engaging with the video game application **302** via the user device **382**). The first computing platform **202** can communicate the first game state data **306** to the game services server **210** (e.g., automatically at predetermined intervals, or in response to a request). The first game state data **306** (or telemetry data **306**) can include any state or telemetry data associated with the first video game application **302** and/or execution of the first video game application **302**. In some examples, the first game state data **306** can include data indicating a latency of execution of the first video game application **302** with respect to the first game platform server **206**.

[0043] The game services server **210**, and specifically, the GACS **204**, can receive second game state data **356** from the second computing platform **204**. In particular, the second video game application **352** can be executing at the second computing platform **204** (e.g., a user **390** is engaging with the video game application **302** via the user device **392**). The second computing platform **204** can communicate the second game state data **356** to the game services server **210** (e.g., automatically at predetermined intervals, or in response to a request). The second game state data **356** (or telemetry data **356**) can include any state or telemetry data associated with the second video game application **352** and/or execution of the second video game application **352**. In some examples, the second game state data **356** can include data indicating a latency of execution of the second video game application **352** with respect to the second game platform server **208**.

[0044] The game services server **210**, and specifically, the GACS **304**, can receive a request to display at the first computing platform **202** a first communication interface **310** of the GACS **304** at the GUI **308** of the first computing platform **202**. That is, the first communication interface **310** is an instance (at least partially) of the GACS **304** at the first computing platform **202**. That is, the first computing platform **202** can instantiate the instance of the GACS **304** at the first computing platform **202** as the first communication interface **310**. The first communication interface **310** can include interactable graphical elements **320**. The interactable graphical elements **320** can include graphical elements that allow or enable user interaction with them—e.g., through control input through the user device **382**.

[0045] The first computing platform **202** receives user input. In some examples, the first computing platform **202** receives the user input from the user **380** via the user device **382**. The user input can indicate a surfacing of the first communication interface **310** at the GUI **308**. FIG. **4**A illustrates the GUI **308** including the first communication interface **310** surfaced at the GUI **308**. Additionally, the GUI **380** illustrates that the user **380** has provided additional user input (via the user device **382**) selecting the user **390** ("Bob") from the list of users **404** socially connected to the user **380**.

[0046] Referring back to FIG. **3**, the GACS **304** can generate content for the first communication interface **310**. In some examples, the GACS **304** can generate the content in response to receiving the user input indicating the surfacing of the first communication interface **310**. The GACS **304** can generate the content based on the first state data **306** and the second state data **356**. Specifically, the GACS **304** can generate the content to be displayed at the first computing platform **202** to include communication messages **322** corresponding to each of the interactable graphical elements **320**. In some examples, the GACS **304** can generate the communications messages **322** based on the first state data **306** and/or the second state data **356**. Additionally, the GACS **304** can generate the communication messages **322** based on one or more parameters. For example, the parameters can include a type of the first computing platform **202** and/or a type of the second computing platform **204**; a type of the first video game application **302** (racing, sports, trivia, etc.); an online status of other users socially connected to the user **380** (e.g., through the GACS **304**), including, for example, an online status of the user **390**; the relationship status of the others users socially connected to the user **380** through the GACS **304**, including, for example, the relationship status of the user **390**; an interaction history of the user **380** with the other users socially connected to the user **380**, including for example, the interaction history between the user **380** and the user **390**; a recent interaction history of the user **380** with the other users socially connected to the user **380**, including, for example, a recent interaction history between the user **380** and the user **390**; a video game application currently being played by the other users socially connected with the user **380** including, for example, the video game application the user **390** is currently playing; among other parameters. In some examples, the GACS **304** can generate the communication messages **322** based on a message history of the user **380** with respect to the other users socially connected to the user **380**, including, for example the user **390**. In some examples, the communication messages **322** can be agnostic to the first video game application **302**, the second video game application **352**, or both.

[0047] In some examples, the GACS **304** can generate the communications messages **322** based on the first state data **306** and the second state data **356** utilizing machine learning (ML), and/or using a neural network.

[0048] The GACS **304** can cause to display the generated content via the first communication interface **310** of the first computing platform **302**. That is, the GACS **304** can cause to display the communication messages **322** at the first communication interface **310** of the first computing platform **302**. Specifically, the GACS **304** can populate the first communication interface **310** with the communication messages **322**. FIG. **4**B illustrates the first communication interface **310** populated with the communication messages **322**.

[0049] Referring back to FIG. **3**, the GACS **304** receives data indicating a selection of one of the communication messages **322** displayed via the first communication interface **310** of the first computing platform **202** to transmit to the second computing platform **204**. Specifically, the GACS **304** receives further user input through the first communication interface **310**. In some examples, the first computing platform **202** receives the further user input from the user **380** via the user device **382**. The user input can indicate a selection of a particular communication message **322** of the communication messages **322**. FIG. **4**B illustrates the first communication interface **310** indicating the selection of the particular communication message **408** of the communication messages **322**. For example, the particular communication message **408** indicates "GG."

[0050] The game services server **210**, and in particular, the GACS **304**, transmits the particular communication message **322** to the second computing platform **204**. For example, the game services server **210**, and in particular, the GACS **304**, transmits the particular communication message **408** to the second computing platform **204**. FIG. **4**C illustrates the first communication interface **310** indicating that the particular communication message **408** (shown in FIG. **4**B) was transmitted to the second computing platform **204**—shown by the element **412** ("Sent").

[0051] Referring back to FIG. **3**, the second computing platform **204** receives the particular communication message **322** from the game services server **210**, and in particular, the GACS **304**. For example, the second computing platform **204** receives the particular communication message **408**.

[0052] The game services server **210**, and specifically, the GACS **304**, can cause to display at the second computing platform **204** a second communication interface **360** of the GACS **304** at the GUI **358** of the second computing platform **204**. That is, the second communication interface **360** is an instance (at least partially) of the GACS **304** at the second computing platform **204**. That is, the second computing platform **204** can instantiate the instance of the GACS **304** at the second computing platform **204** as the second communication interface **360**.

[0053] The second communication interface **360** can include interactable graphical elements **370**. The interactable graphical elements **370** are graphical elements that allow or enable user interaction with them—e.g., through control input through the user device **382**.

[0054] To that end, the GACS **320** can generate the second communication interface **360** to include the particular communication message **322**. Specifically, the GACS **320** can overlay the second communication interface **360** in the foreground of the GUI **358** of the second computing platform **204**. That is, the GUI **358** can include an online game interface, and the GACS **320** can overlay the second communication interface **360** in the foreground of the GUI **358** while concurrently displaying the online game interface in the background of the GUI **358**. FIG. **5**A illustrates the GUI **358** of the second computing platform **204** displaying an online game interface **364** in the background of the GUI **358** with the second communication interface **360** overlayed over the online game interface **364** and displayed in the foreground of the GUI **358** (i.e., the second communication interface **360** is "layered" on top of the online game interface **364**). In particular, the second communication interface **360** includes the particular communication message **408** (e.g., "GG"). The second communication interface **360** can further indicate a source **502** (e.g., the connected user) of the particular communication message **408**.

[0055] Referring back to FIG. **3**, in some examples, the GACS **304** can translate, after receiving the particular communication message **322**, the particular communication message **322** from a first format associated with the first computing platform **202** to a second format associated with the second computing platform **204**. The GACS **304** can generate the second communication interface **360** to include the particular communication message **322** in the translated form (e.g., the particular communication message in the second format). In some examples, the first format of the first computing platform **202** can be based on preferences of the user **380** (user settings of the first computing platform **202**), and the second format of the second computing platform **204** can based on preferences of the user **390** (user settings of the second computing platform **204**).

[0056] In some examples, the GACS **304** can translate the communications messages **322** utilizing machine learning (ML), and/or using a neural network.

[0057] In some examples, the user settings of the first computing platform **202** can include first language settings; and the user settings of the second computing platform **204** can include second language settings (differing form the first language settings). For example, the first format of the first computing platform **202** can include a particular language, or dialect, and the second format of the second computing platform **204** can include a differing language, or different dialect than that of the first format. The GACS **304** can translate the language of the first format of the particular communication message to the language of the second format that is associated with the second computing platform **204**.

[0058] In some examples, the user settings of the first computing platform **202** can include first accessibility settings; and the user settings of the second computing platform **204** can include second accessibility settings (differing from the first accessibility settings). For example, the accessibility settings can include vision-based settings. For example, the second user **390** may have low vision, and the second user **390** can adjust the second accessibility settings associated with the second computing platform **204** based on such—the second user **390** can adjust the second accessibility settings to indicate translation of text-based messages (the particular communication message) to icon-based messages.

[0059] In response to the GACS **320** generating the second communication interface **360** to include the particular communication message **322**, the second computing platform **204** receives user input. In some examples, the second computing platform **204** receives the user input from the user **390** via the user device **392**. The user input can indicate a reply to the particular communication message **322**. For example, the user input can include an interaction with the second communication interface **360**—the user **390** providing input at/via the second communication interface **360**.

[0060] The GACS **304** can generate further content for the second communication interface **360**. In some examples, the GACS **304** can generate the further content in response to receiving the user input indicating the reply to the particular communication message **322**.

[0061] The GACS **304** can generate the further content based on the first state data **306** and the second state data **356**. Specifically, the GACS **304** can generate the further content to be displayed at the second computing platform **204** to include the communication messages **372** corresponding to each of the interactable elements **370**. In some examples, the GACS **304** can generate the further communications messages **372** based the first state data **306** and the second state data **356**. Additionally, the GACS **304** can generate the further communication messages **372** based on one or more parameters. For example, the parameters can include a type of the first computing platform **202** and/or a type of the second computing platform **204**; a type of the first video game application **302** (racing, sports, trivia, etc.); an online status of the others users socially connected to the user **390** (e.g., through the GACS **304**), including, for example, an online status of the user **380**; the relationship status of the other users socially connected to the user **390** through the GACS **304**, including, for example, the relationship status of the user **380**; an interaction history of the user **390** with the other users socially connected to the user **390**, including for example, the interaction history between the user **390** and the user **380**; a recent interaction history of the user **390** with the others users socially connected to the user **390**, including, for example, a recent interaction history between the user **390** and the user **380**; a video game application currently being played by the other users socially connected with the user **390** including, for example, the video game application the user **380** is currently playing; among other parameters. In some examples, the GACS **304** can generate the further communication messages **372** based on a message history of the user **390** with respect to the other users socially connected to the user **390**, including, for example the user **380**. In some examples, the further communication messages **372** can be agnostic to the first video game application **302**, the second video game application **352**, or both.

[0062] In some examples, the GACS **304** can generate the further communications messages **372** based on the first state data **306** and the second state data **356** utilizing machine learning (ML), and/or using a neural network.

[0063] The GACS **304** can cause to display the generated further content via the second communication interface **360** of the second computing platform **204**. That is, the GACS **304** can cause to display the further communication messages **372** at the second communication interface **360** of the second computing platform **204**. Specifically, the GACS **304** can expand the second communication interface **360**. FIG. 5B illustrates the GUI **358** including the second communication interface **360** in an expanded format (expanded second communication interface **360**). Further, the

GACS **304** can populate the expanded second communication interface **360** with the further communication messages **372**. In some examples, the GACS **304** expands the second communication interface **360** and populates the second communication interface **360** with the further communication messages **372** in response to receiving user input from the user **390** via the user device **392**.

[0064] Referring back to FIG. **3**, the GACS **304** receives data indicating a selection of one of the further communication messages **372** via the second communication interface **360** of the second computing platform **304** to transmit to the first computing platform **202**. Specifically, the GACS **304** receives user input through the expanded second communication interface **360**. In some examples, the second computing platform **204** receives the further user input from the user **390** via the user device **392**. The user input can indicate a selection of a further communication message **372** of the further communication messages **372**. FIG. **5**C illustrates the second communication interface **360** indicating the selection of the further particular communication message **506** of the further communication messages **372**. For example, the further particular communication message **506** indicates "GG."

[0065] Referring back to FIG. **3**, the game services server **210**, and in particular, the GACS **304**, transmits the further particular communication message **372** to the first computing platform **202**. For example, the game services server **210**, and in particular, the GACS **304**, transmits the further particular communication message **372** to the first computing platform **202**.

[0066] The first computing platform **202** receives the further particular communication message **372** from the game services server **210**, and in particular the GACS **304**. For example, the first computing platform **202** receives the further particular communication message **506**.

[0067] The GACS **304** can update the first communication interface **310** to include the further particular communication message. In some examples, the GACS **304** updates the first communication interface **310** in response to receiving the selection of the further particular communication message **372** from the second computing platform **204**. Specifically, the GACS **304** can update the first communication interface **310** to overlay the first communication interface **310** in the foreground of the GUI **308** of the first computing platform **202** (or a portion of the first communication interface **310** to overlay in the foreground of the GUI **308**). That is, in some examples, the GUI **308** can include an online game interface. The GACS **304** can overlay the first communication interface **310** in the foreground of the GUI **308** while concurrently displaying the online game interface in the background of the GUI **308**.

[0068] In some examples, the GACS **304** updates the second communication interface **360** to include the further particular communication message **372**. For example, referring to FIG. **5**B, the second communication interface **360** is updated to include the further particular communication message **506**.

[0069] To that end, by having the second communication interface **360** overlaying the online game interface **364**, the user **390** is able to receive and reply to messages (notifications, chats) without having to open a full screen communication interface at the GUI **358**. This can provide a more efficient use of memory of the second computing platform **204**, e.g., during gameplay of the first online game module **354** through the online game interface **364**.

[0070] As noted above, the GACS **304** can generate the communication messages **322** for display on the first communication interface **310** at the first computing platform **202**. Specifically, referring to FIG. **4**B, the first communication interface **310** can include an element **420**, such that when "selected" by the user **380** via the user device **382**, the GACS **304** can update the first communication interface **310**, as shown in FIG. **6**, to include the additional communication messages **602**. The GACS **304** can receive the further user input indicating a selection of the particular communication message of the additional communication messages **602**.

[0071] In some examples, the particular communication message **322** that is transmitted to the second computing platform **204** (by the GACS **304**) can include a function (a computer-implemented action) associated with the first video game application **302** or the second video game application **352**. For example, the function can include an invitation of joining interaction with the first video game application **302** and the user **380** at the second competing platform **204**. For example, the functionality of inviting the user **390** at the second computing platform **204** to join the user **380** in a video game (e.g., the first video game application **302**) can be implemented within the particular communication message **322**. This provides efficient and effective transmission of invitations between the first computing platform **202** and the second computing platform **204** (through the GACS **304**).

[0072] The GACS **304** can provide the particular communication message **322** to the second computing platform **204**, as described above, with the particular communication message **322** including the function associated with a computer-implemented action of the first video game application **302**.

[0073] The second computing platform **204** can receive the particular communication message **322** from the GACS **304**, as described herein. In response to receiving the particular communication message **322** with the function, the GACS **304** can update the second communication interface **360** to include the particular communication message **322** with the function. In particular, the particular communication message **322** that is displayed at the second communication interface **360** can include the functionality indicating their acceptance to join interaction with the second video game application **352** at the second computing platform **204** and functionality indicating a declination to join interaction with the second video game application **352** at the second computing platform **204**. For example, the particular communication message **322** can include text such as “Invite to Join Party” with a first functionality indicating acceptance of the invite (e.g., “yes”) and a second functionality indicating declination of the invite (e.g., “no”).

[0074] To that end, the functionality of the invitation of joining interaction with the first video game application **302** can be independent, or agnostic, of the first video game application **302**. For example, the second computing platform **204** can be implementing/executing a differing video game application than the first video game application **302**, and receive the particular communication message including the function of the invitation of joining interaction with the first video game application **302**.

[0075] In some further examples, the function associated with the first video game application **302** or the second video game application **352** can include joining or leaving a party (e.g., an online multi-person cooperative session of the first video game application **302** or the second video game application **352**). In some further examples, the function associated with the first video game application **302** or the second video game application **352** can include providing, accepting, or rejecting a social connection request (e.g., “friend request”) between the users **380**, **390** through the GACS **304**. In some further examples, the function associated with the first video game application **302** or the second video game application **352** can include queuing up for a particular game mode of the video game application **302** or the second video game application **352**. In some further examples, the function associated with the first video game application **302** or the second video game application **352** can include leading a particular game of the video game application **302** or the second video game application **352**. In some further examples, the function associated with the first video game application **302** or the second video game application **352** can include joining a voice chat associated with the first video game application **302** or the second video game application **352**. In some further examples, the function associated with the first video game application **302** or the second video game application **352** can include leaving a party associated with the first video game application **302** or the second video game application **352**.

[0076] FIG. **7** illustrates a flowchart depicting selected elements of an embodiment of a method **700** for facilitating communication between computing platforms. The method **700** may be performed by the computing platform **100**, the first computing platform **202**, the second computing platform **204**, and/or the game services server **210**, and with reference to FIGS. **1-6**. It is noted that certain operations described in method **700** may be optional or may be rearranged in different embodiments.

[0077] The GACS **304** establishes, over the network **212**, communication with the first computing platform **202** and the second computing platform **204**, at **702**. The GACS **304** receives the first game state data **306** associated with the first video game application **302** executing at the first computing platform **202**, at **704**. The GACS **304** receives the second game state data **356** associated with the second video game application **352** executing at the second computing platform **204**, at **706**. The GACS **304** receives a request to display at the first computing platform **202** the first communication interface **310** of the GACS **304**, at **708**. The first communication interface **310** includes interactable graphical elements **320**. The GACS **304** generates content for the first communication interface **310** to be displayed at the first computing platform **202**, at **710**. The content is generated base at least in part on the first game state data **306** or the second game state data **356**. The content to be displayed includes the communication messages **322** corresponding to each of the interactable graphical elements **320**. Each of the communication messages **322** comprises a function associated with the first video game application **302** or the second video game application **352**. The GACS **304** causes to display the generated content via the first communication interface **310** of the first computing platform **202**, at **712**. The GACS **304** receives a selection of one of the communication messages **322** displayed via the first communication interface **310** of the first computing platform **202** to the transmit to the second computing platform **204**, at **714**. The GACS **304** transmits the selected communication message **322** to the second computing platform **204** for display via the second communication interface **360** of the GACS **304** at the second computing platform **204**, at **716**.

[0078] The above disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other embodiments which fall within the true spirit and scope of the present disclosure. Thus, to the maximum extent allowed by law, the scope of the present disclosure is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing detailed description.

[0079] Herein, “or” is inclusive and not exclusive, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A or B” means “A, B, or both,” unless expressly indicated otherwise or indicated otherwise by context. Moreover, “and” is both joint and several, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A and B” means “A and B, jointly or severally,” unless expressly indicated otherwise or indicated other-wise by context.

[0080] The scope of this disclosure encompasses all changes, substitutions, variations, alterations, and modifications to the example embodiments described or illustrated herein that a person having ordinary skill in the art would comprehend. The scope of this disclosure is not limited to the example embodiments described or illustrated herein. Moreover, although this disclosure describes and illustrates respective embodiments herein as including particular components, elements, features, functions, operations, or steps, any of these embodiments may include any combination or permutation of any of the components, elements, features, functions, operations, or steps described or illustrated anywhere herein that a person having ordinary skill in the art would comprehend. Furthermore, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative.

[0081] It should be understood that the original applicant herein determines which technologies to use and/or productize based on their usefulness and relevance in a constantly evolving field, and what is best for it and its players and users. Accordingly, it may be the case that the systems and methods described herein have not yet been and/or will not later be used and/or productized by the original applicant. It should also be understood that implementation and use, if any, by the original applicant, of the systems and methods described herein are performed in accordance with its privacy policies. These policies are intended to respect and prioritize player privacy, and to meet or exceed government and legal requirements of respective jurisdictions. To the extent that such an implementation or use of these systems and methods enables or requires processing of user personal information, such processing is performed (i) as outlined in the privacy policies; (ii) pursuant to a valid legal mechanism, including but not limited to providing adequate notice or where required, obtaining the consent of the respective user; and (iii) in accordance with the player or user's privacy settings or preferences. It should also be understood that the original applicant intends that the systems and methods described herein, if implemented or used by other entities, be in compliance with privacy policies and practices that are consistent with its objective to respect players and user privacy.

Claims

1. A computer-implemented method of facilitating communication between computing platforms, the method including: establishing, over a network and via a game agnostic communication service, communication with a first computing platform and a second computing platform; receiving first game state data associated with a first video game application executing at the first computing platform; receiving second game state data associated with a second video game application executing at the second computing platform; receiving a request to display at the first computing platform a first communication interface of the game agnostic communication service, the first communication interface including one or more interactable graphical elements; generating content for the first communication interface to be displayed at the first computing platform, the content being generated based at least in part on the first game state data or the second game state data, wherein the content to be displayed includes communication messages corresponding to each of the one or more interactable graphical elements, wherein each of the communication messages comprise a function associated with the first video game application or the second video game application; causing to display the generated content via the first communication interface of the first computing platform; receiving a selection of one of the communication messages displayed via the first communication interface of the first computing platform to transmit to the second computing platform; and transmitting the selected communication message to the second computing platform for display via a second communication interface of the game agnostic communication service at the second computing platform.
2. The computer-implemented method of claim 1, wherein generating the content to be displayed is based at least in part on the type of platforms of the first computing platform and second computing platform.
3. The computer-implemented method of claim 1, wherein both the first video game application and the second video game application are of the same video game application.
4. The computer-implemented method of claim 1, wherein the first video game application is of a first type of video game application and the second video game application is of a second type of video game application differing from the first type of video game application.
5. The computer-implemented method of claim 1, further including: translating the selected communication message from a first format associated with the first computing platform to a second format associated with the second computing platform; and displaying the translated selected communication message via the second communication interface of the second computing platform.
6. The computer-implemented method of claim 5, wherein translating the selected communication

message includes translating the selected communication message based on one or more user settings of the second computing platform.
7. The computer-implemented method of claim 6, wherein the user settings include accessibility settings, language settings, or both.
8. The computer-implemented method of claim 1, wherein generating the content to be displayed is also based on one or more settings from at least the first video game application or the second video game application.
9. The computer-implemented method of claim 1, wherein the first computing platform is of a first type of gaming platform and the second computing platform is of a second type of gaming platform, the first type of gaming platform differing from the second type of gaming platform.
10. The computer-implemented method of claim 1, further including instantiating the first communication interface at the first computing platform by the first video game application.
11. A server computing system comprising a processor having access to memory media storing instructions executable by the processor to perform operations: establishing, over a network and via a game agnostic communication service, communication with a first computing platform and a second computing platform; receiving first game state data associated with a first video game application executing at the first computing platform; receiving second game state data associated with a second video game application executing at the second computing platform; receiving a request to display at the first computing platform a first communication interface of the game agnostic communication service, the first communication interface including one or more interactable graphical elements; generating content for the first communication interface to be displayed at the first computing platform, the content being generated based at least in part on the first game state data or the second game state data, wherein the content to be displayed includes communication messages corresponding to each of the one or more interactable graphical elements, wherein each of the communication messages comprise a function associated with the first video game application or the second video game application; causing to display the generated content via the first communication interface of the first computing platform; receiving a selection of one of the communication messages displayed via the first communication interface of the first computing platform to transmit to the second computing platform; and transmitting the selected communication message to the second computing platform for display via a second communication interface of the game agnostic communication service at the second computing platform.
12. The server computing system of claim 11, wherein generating the content to be displayed is based at least in part on the type of platforms of the first computing platform and second computing platform.
13. The server computing system of claim 11, wherein both the first video game application and the second video game application are of the same video game application.
14. The server computing system of claim 11, wherein the first video game application is of a first type of video game application and the second video game application is of a second type of video game application differing from the first type of video game application.
15. The server computing system of claim 11, the operations further including: translating the selected communication message from a first format associated with the first computing platform to a second format associated with the second computing platform; and displaying the translated selected communication message via the second communication interface of the second computing platform.
16. The server computing system of claim 15, wherein translating the selected communication message includes translating the selected communication message based on one or more user settings of the second computing platform.
17. The server computing system of claim 16, wherein the user settings include accessibility settings, language settings, or both.
18. The server computing system of claim 11, wherein generating the content to be displayed is

also based on one or more settings from at least the first video game application or the second video game application.

19. The server computing system of claim 11, wherein the first computing platform is of a first type of gaming platform and the second computing platform is of a second type of gaming platform, the first type of gaming platform differing from the second type of gaming platform.

20. A non-transitory computer-readable medium storing software comprising instructions executable by one or more computers which, upon such execution, cause the one or more computers to perform operations comprising: establishing, over a network and via a game agnostic communication service, communication with a first computing platform and a second computing platform; receiving first game state data associated with a first video game application executing at the first computing platform; receiving second game state data associated with a second video game application executing at the second computing platform; receiving a request to display at the first computing platform a first communication interface of the game agnostic communication service, the first communication interface including one or more interactable graphical elements; generating content for the first communication interface to be displayed at the first computing platform, the content being generated based at least in part on the first game state data or the second game state data, wherein the content to be displayed includes communication messages corresponding to each of the one or more interactable graphical elements, wherein each of the communication messages comprise a function associated with the first video game application or the second video game application; causing to display the generated content via the first communication interface of the first computing platform; receiving a selection of one of the communication messages displayed via the first communication interface of the first computing platform to transmit to the second computing platform; and transmitting the selected communication message to the second computing platform for display via a second communication interface of the game agnostic communication service at the second computing platform.
